

Limit of function

$$\frac{3}{0} \rightarrow \text{undefined}$$

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow a} f(x) =$$

$$y = \frac{1}{x-3}, \quad \frac{1}{5-3} = \frac{1}{2} \text{ undefined} \quad D: \mathbb{R} - \{3\}, \quad R: \mathbb{R} - \{f(3)\}$$

$$\lim_{x \rightarrow 3} \frac{1}{x-3} =$$

Two sided limit

$$L.H.L = \text{left hand limit} = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$$

$$R.H.L = \text{right hand limit} = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right) = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ Does not exist}$$

$$y = f(x) = \frac{1}{x}$$

$$D: \mathbb{R} - \{0\}, \quad \boxed{x=0}$$

$$R: \mathbb{R} - \{0\}, \quad \boxed{y=0}$$

$$y = \frac{1}{x^2}, \quad \mathbb{R} - \{0\}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x^2} = +\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

$$\lim_{x \rightarrow a} f(x) \neq f(a)$$

Defⁿ: If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a), then we write

$$\lim_{x \rightarrow a} f(x) = L$$

which we read, "the limit of $f(x)$ as x approaches to a is L ".

$$\text{Ex: Let } f(x) = \frac{|x|}{x}, \text{ find } \lim_{x \rightarrow 0} f(x).$$

$$\text{Sol: } L.H.L = \lim_{x \rightarrow 0^-} f(x), \quad R.H.L = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{|-h|}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h}{-h}$$

$$= \boxed{-1}$$

$$= \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{|h|}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h}$$

$$= \boxed{1}$$

$$\lim_{x \rightarrow 0} f(x) \text{ Does not exist}$$

$$1.5 \neq 1.4$$

$$y = f(x) = \frac{|x|}{x} = \begin{cases} \frac{x}{x}, & x > 0 \\ -\frac{x}{x}, & x < 0 \end{cases} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

